Page 1 | Sources, definitions and base period

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Saved-source extracts below are filing facts, not forecast assumptions. All money is USD millions. The full source ledger and raw snapshots remain in the repository.

SEC tag / value	Period / publication / accession
CashAndCashEquivalentsAtCarryingValue = 39,544.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020
LongTermDebtCurrent = 11,007.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020
LongTermDebtNoncurrent = 71,340.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020
StockholdersEquity = 107,520.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020
Assets = 383,266.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020
Liabilities = 275,746.0 USD millions	End 2026-06-27; filed 2026-07-31; accession 0000320193-26-000020

Source: data/fact_ledger.json; Apple June 2026 Form 10-Q, filed July 31, 2026.

TTM basis: FY2025 + current nine months - prior nine months. June published balances are opening proxies for five annual decision periods ending September 2027-2031; no September actual balances or unreported FY2026 annual results are invented.

Facts: reported balances and historical income/cash flows. Normalization: Products margin reserve to 38%, not a measured refund adjustment. Forecasts: category growth, margins, investment, tax and discount assumptions. Outputs: linked statements and valuation.

Page 2 | Operating forecast and checks that fail

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Actual model.py excerpts. Category forecasts feed earnings, cash flows and the balance sheet; accounting checks are executed for every forecast row. Omitted lines remain available in the source file.

```
 61  def project(a):
 62      validate_assumptions(a); base_reconciliation(); p=copy.deepcopy(H['2026']); rows=[]
 63      for i in range(5):
 64          r={k:p[k] for k in ASSETS+LIABS+('equity',)}; r['year']=2027+i
 65          for k in ['iphone','mac','ipad','wearables','services']: r[k]=p[k]*(1+a[k+'_growth'][i])
 66          r['products']=sum(r[k] for k in ['iphone','mac','ipad','wearables']);
     r['revenue']=r['products']+r['services']
 67          r['products_gp']=r['products']*a['product_margin'];
     r['services_gp']=r['services']*a['services_margin']
 68          r['gross_profit']=r['products_gp']+r['services_gp']; r['cogs']=r['revenue']-r['gross_profit']
 69          r['rd']=r['revenue']*a['rd_ratio']; r['sga']=r['revenue']*a['sga_ratio']
 70          r['ebit']=r['gross_profit']-r['rd']-r['sga'] # D&A and SBC already in margins/opex.
 71          r['net_other']=a['net_other']; r['pretax']=r['ebit']+r['net_other']
 72          r['tax']=max(r['pretax'],0)*a['tax']; r['ni']=r['pretax']-r['tax']
     ... [intervening source lines omitted]
 81          r['delta_wc']=wc(r)-wc(p); r['sbc']=r['revenue']*a['sbc_ratio']
 82          r['cfo']=r['ni']+r['da']+r['sbc']-r['delta_wc']
 83          r['cfi']=-r['capex']-r['other_reinvestment']
     ... [intervening source lines omitted]
 87          r['opening_cash']=p['cash']; r['cash']=p['cash']+r['cfo']+r['cfi']+r['cff']
 88          r['equity']=p['equity']+r['ni']+r['sbc']-r['dividends']-r['buybacks']
 89          r['nopat']=r['ebit']*(1-a['tax'])
 90          # SBC treated as an economic cost: its CFO addback is reversed in FCFF.
 91          r['fcff']=r['nopat']+r['da']-r['capex']-r['other_reinvestment']-r['delta_wc']
 92          r['fcff_from_cfo']=r['cfo']-r['sbc']-r['net_other']*(1-a['tax'])-r['capex']-r['other_reinvestment']
 93          r['assets']=sum(r[k] for k in ASSETS); r['liabilities']=sum(r[k] for k in LIABS)
 94          check_row(r,a); rows.append(r); p=r
 95      return rows
 96  def check_row(r,a):
 97      if not all(math.isfinite(v) for v in r.values()): raise ValueError('Nonfinite forecast')
 98      near(balance(r),0,f"FY{r['year']} balance sheet")
 99      near(r['cash'],r['opening_cash']+r['cfo']+r['cfi']+r['cff'],'Forecast cash tie')
100      near(r['fcff'],r['fcff_from_cfo'],'Operating FCFF versus cash-flow bridge')
101      if r['cash']<a['cash_floor']-1e-5: raise ValueError('Cash-floor breach: funding assumption
     insufficient')
102      if any(r[k]<-1e-5 for k in ASSETS+LIABS): raise ValueError('Negative asset or liability')
```

Real correction: an omitted separately presented intangible asset of 20,342m caused a balance failure. The filing supplied the missing line; aggregation is explicit. No balancing plug was added.

Page 3 | DCF, funded growth and equity bridge


```
103  def dcf_path(path,k,g,cash,debt,shares,terminal_fcff=None):
104      if len(path)!=5 or not all(math.isfinite(v) for v in path): raise ValueError('Five finite FCFF values
     required')
105      if not all(math.isfinite(v) for v in (k,g,cash,debt,shares)) or k<=g or k<=0 or g<0 or shares<=0 or
     debt<0: raise ValueError('Invalid DCF boundary or bridge')
106      terminal_fcff=path[-1]*(1+g) if terminal_fcff is None else terminal_fcff
107      if not math.isfinite(terminal_fcff) or terminal_fcff<=0: raise ValueError('Nonpositive terminal FCFF:
     perpetuity unresolved')
108      explicit=sum(v/(1+k)**i for i,v in enumerate(path,1)); tv=terminal_fcff/(k-g); pvt=tv/(1+k)**5
109      ev=explicit+pvt; equity=ev+cash-debt
110      near(equity-ev,cash-debt,'EV to equity bridge')
111      return dict(fcff=path,pv_explicit=explicit,terminal_fcff=terminal_fcff,terminal_value=tv,pv_terminal=pv
     t,ev=ev,nonoperating_assets=cash,debt=debt,equity=equity,shares=shares,per_share=equity/shares,terminal_sha
     re=pvt/ev)
112  def value(rows,a):
113      for r in rows: check_row(r,a)
114      h=H['2026']; cash=h['cash']+h['securities_current']+h['securities_long']-a['cash_floor']
115      debt=h['commercial_paper']+h['debt_current']+h['debt_long']
116      # Terminal growth requires total reinvestment at an explicit incremental ROIC.
117      nopat6=rows[-1]['nopat']*(1+a['terminal_growth'])
118      term=nopat6*(1-a['terminal_growth']/a['terminal_roic'])
119      result=dcf_path([r['fcff'] for r in
     rows],a['wacc'],a['terminal_growth'],cash,debt,M['valuation_shares'],term)
120      result['terminal_nopat']=nopat6; result['terminal_reinvestment']=nopat6-term
```

FCFF is operating cash available to capital providers. Financing distributions and debt repayments are excluded. Stock compensation remains an economic expense; depreciation is not deducted twice.

Core terminal convention: next-period NOPAT multiplied by (1 - growth / incremental ROIC), then divided by WACC minus growth. Terminal growth therefore consumes reinvestment.

Separate duration_challenge test: first five operating periods stay fixed; additional growth fades toward the mature rate, with reinvestment funded. A longer growth runway is an economic assumption, not something proven by a product release.

Base bridge, USD millions: EV 2,122,692 + liquid assets 146,517 - operating reserve 30,000 - debt 84,344 = equity 2,154,865; divided by 14,649.596m diluted-proxy shares gives approximately $147.09/share.

Page 4 | Actual test execution and cold run

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Fresh actual test output captured while assembling this packet. This is saved execution evidence, not a live terminal display.

```
test_existing_prediction_cannot_be_overwritten
  (test_evidence_tools.EvidenceToolTests.test_existing_prediction_cannot_be_overwritten) ... ok
test_nonfinite_change_rejected (test_evidence_tools.EvidenceToolTests.test_nonfinite_change_rejected) ... ok
test_prepare_records_without_execution
  (test_evidence_tools.EvidenceToolTests.test_prepare_records_without_execution) ... ok
test_provenance_hashes_cover_model_inputs_and_helper
  (test_evidence_tools.EvidenceToolTests.test_provenance_hashes_cover_model_inputs_and_helper) ... ok
test_articulation_all_cases (test_model.RefreshedTests.test_articulation_all_cases) ... ok
test_boundaries_and_negative_explicit (test_model.RefreshedTests.test_boundaries_and_negative_explicit) ... ok
test_bridge (test_model.RefreshedTests.test_bridge) ... ok
test_fail_loudly (test_model.RefreshedTests.test_fail_loudly) ... ok
test_growth_duration_terminal_reconciliation
  (test_model.RefreshedTests.test_growth_duration_terminal_reconciliation) ... ok
test_historical_independent_filing (test_model.RefreshedTests.test_historical_independent_filing) ... ok
test_judgment_and_output_are_separate (test_model.RefreshedTests.test_judgment_and_output_are_separate) ... ok
test_monotonicity (test_model.RefreshedTests.test_monotonicity) ... ok
test_normalization (test_model.RefreshedTests.test_normalization) ... ok
test_operating_direction_before_run_expectation
  (test_model.RefreshedTests.test_operating_direction_before_run_expectation) ... ok
test_peers_and_reverse (test_model.RefreshedTests.test_peers_and_reverse) ... ok
test_source_cutoff_and_hashes (test_model.RefreshedTests.test_source_cutoff_and_hashes) ... ok
test_training_answer (test_model.RefreshedTests.test_training_answer) ... ok
----------------------------------------------------------------------
Ran 17 tests in 0.079s
OK
```

Command: python -m unittest discover -s . -p "test_*.py" -v

Separate cold-run record: fresh isolated standard-library environment; tests and end-to-end output generation passed; generated results exactly matched frozen results.json. Full log: docs/cold-run-log.txt.

A passing test suite validates calculation and accounting behavior; it cannot establish that AI adoption or a long growth runway will materialize.

Page 5 | Your prediction preserved before execution

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Human statement, preserved verbatim before this scenario was executed:

I want to increase the original fcff growth estimate, change it from .57% to 1%, valueaiton should rise, maybe by a factor of 1.5, and no since my original recommendation is to buy i stick with that

Field	Preserved prediction
Input / units	Annual FCFF growth in each of five explicit years; 0.57% -> 1.00% (+0.43 percentage points)
Expected direction / size	Higher valuation; about 1.5 times original value (+50%)
Expected decision	Buy / initiate unchanged
Recorded timestamp UTC	2026-10-16T20:40:37.195446+00:00
Pre-run Git commit	4173681346adb408ed11ff4c2160809a112b9eab
Commit timestamp	2026-10-16T16:40:37-04:00

Scope: unchanged original repository-root dcf.py, FY2025 flat-FCFF exercise. This is separate from the September2026 operating-driven forecast.

Provenance: Alex supplied the prediction; the assistant recorded, committed and executed it under authorization. No AI-off prediction declaration or human-only execution is asserted. The preserved original prediction file is unchanged.

Page 6 | Actual result, diagnosis and limits

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Metric	Before -> after
Annual explicit FCFF growth	0.57% -> 1.00%
Per-share valuation	$90.6619 -> $92.3321
Year5 FCFF, USD millions	98,378.47 -> 100,499.68
Actual valuation ratio / change	1.018423x / +1.8423%
Prediction comparison	Direction correct; size much smaller than predicted 1.5x

Alex's actual post-run response:

my size estimate differed from the result because I did not do the exact math, and I still recommend buy.

Technical explanation supplied by the assistant: the annual explicit growth increment is only 0.43 percentage points for five years. Terminal growth, WACC, original cash/debt and shares are unchanged, so this does not multiply the entire valuation by 1.5.

AI-use and limitations: original historical engine only; active pro-forma unchanged. Human prediction and diagnosis are separated from assistant execution. Acceptance of this protocol and historical-engine scope under the written locked requirement is not presumed. A reported manual beta check does not verify the whole WACC.

Source records: alex-growth-precommit.json, alex-growth-run.json, Alex-Growth-Exercise.md and Validation-and-AI-Use.pdf. This packet is recording evidence, not a completed recording or corrected transcript.